

Column Generation for the Split Delivery Vehicle Routing Problem

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Abstract

In this paper we tackle a variation of the Vehicle Routing Problem (VRP) in which each customer can be served by more than one vehicle, each serving a fraction of its demand. This problem is known as the Split Delivery VRP (SDVRP). Due to the potential savings that can be achieved in this way, the SDVRP recently received great attention in the combinatorial optimization community.

We propose a new extended formulation for the problem. We exploit its properties to derive an effective column generation scheme. Our method is compared to the previous ones in the literature from both a theoretical and a computational point of view. In particular, our formulation involves a polynomial number of constraints and flow variables, can be optimized by solving well understood resource constrained shortest path problems and yields a bound which is not dominated by any previous one in the literature.

1 Introduction

In any logistic chain a set of customers requires the delivery of a certain amount of goods, which is carried by vehicles and is stocked at some depot. Over the last decades optimization methods have been developed and applied with success to goods distribution in order to reduce operational costs and improve efficiency.

In its classical version, the Vehicle Routing Problem (VRP) addresses the situation in which a set of customers is served from a central depot by a fleet of identical vehicles and each customer can be visited by at most one vehicle. For each vehicle, a route, which is a sequence of deliveries, has to be generated so that all customers are serviced and the total distance traveled by the fleet is minimized. This problem is more theoretical than practical but it has been widely studied by researchers. We address the reader to the book of Toth and Vigo (2002) for a comprehensive review of the problem and solution methods.

Despite its modeling simplicity the VRP is a combinatorial problem which is hard to solve. State-of-the-art exact algorithms are able to solve nowadays instances with up to 135 customers (Fukasawa et al., 2006), which justifies the variety of heuristic and meta-heuristic approaches presented so far, see for instance Gendreau et al. (1994).

In the split delivery vehicle routing problem (SDVRP), the constraint on the single visit per customer is relaxed, thus a customer can be serviced by more than one vehicle. We refer to those customers as *split customers*. The fact that the customer's demand can be split among several vehicles allows to better exploit the fleet resources. Potential savings can be achieved by using fewer vehicles and reducing the total distance traveled.

The SDVRP has been introduced by Dror and Trudeau (1989), in a successive work the authors derive some properties of the optimal solution and devise valid inequalities (Dror et al., 1994). The computational complexity of the SDVRP has been studied in Archetti et al. (2005) and a tight bound on the potential savings has been presented in Archetti, Savelsbergh and Speranza (2006b).

Archetti, Speranza and Hertz (2006) propose a tabu search algorithm, which is extended in Archetti, Savelsbergh and Speranza (2006a) to an optimization-based heuristic. In a recent

paper Chen et al. (2007) review the state of the art and present construction and improvement heuristics.

While these heuristics are able to effectively reach good upper bounds, not many exact methods have been proposed for the SDVRP, mainly due to a lack of good lower bounds. The algorithms of Lee et al. (2006) and Jin et al. (2007) can optimally solve only instances with up to 8 and 25 customers respectively. Further lower bounds have been proposed by Belenguer et al. (2000). The only practical exact methods are proposed by Gueguen (1999) and Gendreau et al. (2006), who address the problem with Time Windows using column generation techniques. Their method has the main drawback of requiring the simultaneous generation of columns and rows.

In this paper we propose a new extended formulation for the SDVRP, together with a column generation scheme. This method allows us to obtain a tight lower bound, for both instances with and without Time Windows, which is not dominated by the one of Gendreau et al. (2006) and is much easier to compute.

In Section 2 we define the SDVRP and we review some of its properties which are exploited in our algorithm. In Section 3 we derive our model and in Section 4 we discuss how to obtain optimal solutions using column generation. In Section 5 we compare the quality of our lower bound with several others from the theoretical point of view, proving dominance relations between them. Finally, in Section 6 we experimentally compare our bound with the one of Gendreau et al. (2006), presenting some computational results.

2 Properties of the SDVRP

We provide here the definition of the problem and recall the main properties of the SDVRP, which extend also to the SDVRPTW. All the proofs can be found in Dror and Trudeau (1990) and Gendreau et al. (2006).

Let $G = (N, A)$ be a directed graph where $N = \{0\} \cup V$ be the set of vertices and A the set of arcs. Vertices $V = \{1, \dots, n\}$ correspond to customers, whereas vertex 0 represents the depot. Customers require the delivery of a positive integer quantity d_i of some commodity. A set K of identical vehicles of capacity Q , located at the depot, is available. Each arc $(i, j) \in A$ is associated with a positive cost c_{ij} . In addition, for the SDVRPTW, a positive traverse time t_{ij} is defined for each arc and a time window $[g_i, h_i]$ and service time s_i , which does not depend on the quantity delivered, are defined for each customer $i \in V$. The aim is to find the route of the vehicles and the fraction of customer demands served by each vehicle which minimize the overall travel cost.

1. The SDVRP is NP-Hard in the strong sense.
2. The potential saving of an optimal SDVRP with respect to the optimal VRP on the same instance, that is the difference between the optimal solution values when deliveries can be split or not, tend to two when n tend to $+\infty$. Remarkably, when the number of vehicles is fixed, the cost saving can be arbitrary large.
3. There exists a solution to the SDVRP in which $|K| = \left\lceil \frac{\sum_i d_i}{Q} \right\rceil$.
4. There exists an optimal solution to the SDVRP in which no two routes have more than one split customer in common. As a corollary, there exists an optimal solution in which each arc $(i, j) \in A$, with $i, j \in V$, appears at most once.
5. The previous theorem can be generalized considering the k -split cycles. In a SDVRP solution, a k -split cycle is a set $S = \{v_1, \dots, v_k\} \subseteq V$ of vertices such that vertices v_t and v_{t+1} are fractionally serviced by vehicle $t = 1, \dots, k-1$ and vertices v_1 and v_k are fractionally serviced by vehicle k . There exists an optimal solution to the SDVRP not containing k -split cycles, for $k \geq 2$.

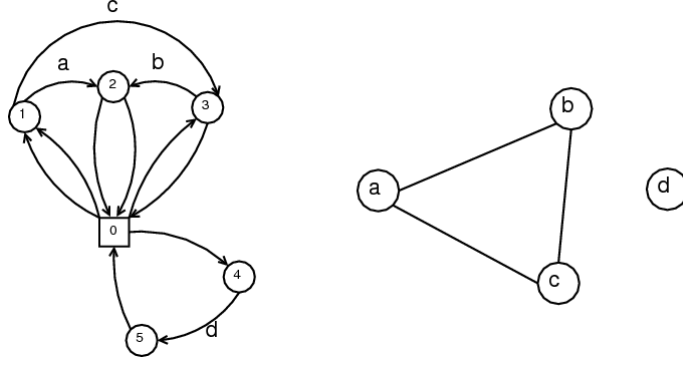


Fig. 1. A k -split cycle over customers 1, 2, and 3 and the associated graph G' , which contains a cycle over routes a, b , and c .

Thanks to Property 4, the number of split customers can be bounded to $|K|$. We can strengthen this bound to $|K| - 1$ as follows. Given a SDVRP solution we define an associated graph $G' = (K, E)$. A node is added for each route k in the solution and an edge e is added among two routes k_i and k_j if they share a split customer. Since two routes share at most one split customer, $|E|$ equals the number of split customers. Any k -split cycle in the associated graph G' is represented as a cycle among k nodes. An illustration of a 3-split cycle is reported in Figure 1. Customers 1 and 2 are served by route a , customers 2 and 3 are served by route b and customers 3 and 1 are served by route c . The associated graph G' has a cycle of length 3 among routes a, b , and c , while route d is represented by a non connected vertex.

Corollary 1. *There exist an optimal solution in which the number of split customers is limited to $|K| - 1$.*

Proof. Since there exists an optimal solution without k -split cycles, the associated graph G' does not contain any cycle of length k . Moreover, since G' has at most $|K|$ nodes, there exists an optimal solution in which the associated graph G' is cycle free. Thus, G' is a tree and any tree with $|K|$ nodes has at most $|E| \leq |K| - 1$ edges.

Through the paper we also assume to handle rational data. This allows us to consider demands and capacities as integers without loss of generality.

3 Formulations

As described in Dror et al. (1994), the SDVRP admits the following three-index flow-based formulation:

$$\min z_{FP} = \sum_{i \in V} \sum_{j \in V} c_{ij} \sum_{k \in K} x_{ijk} \quad (1)$$

$$\text{s.t.} \sum_{i \in V} x_{ilk} = \sum_{j \in V} x_{ljk} \quad \forall l \in V, \forall k \in K \quad (2)$$

$$\sum_{k \in K} y_{ik} = 1 \quad \forall i \in V \quad (3)$$

$$\sum_{i \in V} d_i y_{ik} \leq Q \quad \forall k \in K \quad (4)$$

$$\sum_{j \in V} x_{ijk} \geq y_{ik} \quad \forall i \in V, k \in K \quad (5)$$

$$\text{subtour / connectivity / time windows} \quad (6)$$

$$x_{ijk} \in \{0, 1\} \quad \forall i, j \in V, k \in K \quad (7)$$

$$y_{ik} \geq 0 \quad \forall i \in V, k \in K \quad (8)$$

Each binary variable x_{ijk} takes value 1 if vehicle k visits customer j right after customer i , 0 otherwise; each variable y_{ik} represents the fraction of the demand of customer i served by vehicle k . Besides standard flow conservation constraints (2), constraints (3) impose that each customer is fully served, constraints (4) impose that the capacity of each vehicle is not exceeded, constraints (5) ensure no demand for customer i is served by vehicle k unless vehicle k visits i . We do not explicitly give the formulation of subtour elimination and connectivity constraints, as they deserve a particular attention in split delivery problems (Dror et al., 1994); furthermore, we do not explicitly formulate time windows constraints, as they are naturally treated as side constraints in the remainder. The objective is to minimize overall travel costs.

This formulation involves a polynomial number of variables, but the number of constraints (6) grows exponentially in problem dimension. Tackling the SDVRP by straightly solving this ILP with general purpose solvers is impractical, mainly for two reasons: first, the continuous relaxation of this formulation typically provides a poor bound; moreover, constraints (6) have to be dynamically generated in order to effectively solve the relaxation. Instead, we propose to obtain a lower bound for SDVRP by means of Dantzig-Wolfe reformulation (Nemhauser and Wolsey, 1988). We substitute in the linear relaxation of the flow model (FP), for each $k \in K$, the polyhedron defined by constraints (2), (4), (5) and (6) with the set $X_k = \text{conv}\{(x_{ijk}, y_{ik}) \mid 0 \leq x_{ijk} \leq 1, y_{ik} \geq 0, (2), (4), (5), (6)\}$ that is, the convex hull of its integer points, and we solve the following continuous relaxation of the convexified model:

$$\min z_{FP} = \sum_{i \in V} \sum_{j \in V} c_{ij} \sum_{k \in K} x_{ijk} \quad (9)$$

$$\text{s.t.} \sum_{k \in K} y_{ik} = 1 \quad \forall i \in V \quad (10)$$

$$(x_{ijk}, y_{ik}) \in X_k \quad (11)$$

$$(12)$$

For all $k \in K$, let $\bar{x}^1, \bar{x}^2 \dots$ be the extreme points of X_k ; let Ω_k be the corresponding set of indices. We denote as $\bar{x}_{ij}^r$ and $\bar{y}_i^r$ each component of $\bar{x}^r$. The variables x_{ijk} and y_{ik} can be

expressed as a convex combination of these extreme points:

$$\begin{aligned}
x_{ijk} &= \sum_{r \in \Omega_k} \bar{x}_{ij}^r \lambda_r & \forall i, j \in V \\
y_{ik} &= \sum_{r \in \Omega_k} \bar{y}_i^r \lambda_r & \forall i \in V \\
\text{s.t. } \sum_{r \in \Omega_k} \lambda_r &= 1 \\
\lambda_r &\geq 0 & \forall r \in \Omega_k
\end{aligned}$$

For each $r \in \Omega_k$, we define $c_r = \sum_{i \in V} \sum_{j \in V} c_{ij} \bar{x}_{ij}^r$; by substitution and some standard simplifications, we obtain the following extended formulation:

$$\min z_{CCLP} = \sum_{k \in K} \sum_{r \in \Omega_k} c_r \lambda_r \quad (13)$$

$$\text{s.t. } \sum_{k \in K} \sum_{r \in \Omega_k} \bar{y}_i^r \lambda_r \geq 1 \quad \forall i \in V \quad (14)$$

$$\sum_{r \in \Omega_k} \lambda_r \leq 1 \quad \forall k \in K \quad (15)$$

$$\lambda_r \geq 0 \quad \forall k \in K, r \in \Omega_k \quad (16)$$

Formulation (13) – (16) includes a column for each route and for each combination of demands of customers in the route which does not exceed the capacity of the vehicle. A combination of routes can be selected by setting variables λ_r . Inequalities (15) are convexity constraints, while constraints (14) ensure that the demand of each customer is completely served.

An extreme integer vertex can potentially have at most $|K| - 1$ fractional coordinates, one for each customer served in a fractional way. However, we can exploit some properties of this model to consider far less variables.

Proposition 1. *There exists an optimal solution to (13) – (16) in which the basis is formed only by columns encoding vertices with at most one fractional coordinate.*

We only present a sketch of the proof. Let π_i and μ_k be the dual variables corresponding to constraints (14) and (15) respectively. The reduced cost of each column corresponding to a route $k \in \Omega_k$ is given by $\tilde{c}_r = \sum_{i \in V} \sum_{j \in V} c_{ij} \bar{x}_{ij}^r - \sum_{i \in V} \bar{y}_i^r \pi_i - \mu_k$. Let us consider any optimal column including two fractional coefficients $0 < \bar{y}_i^r < 1$ and $0 < \bar{y}_j^r < 1$. We can identify two corresponding feasible columns as follows. In the first one the coefficient $\bar{y}_i^r$ is increased and the coefficient $\bar{y}_j^r$ is decreased, keeping the same capacity consumption, until either $\bar{y}_i^r$ is 1 or $\bar{y}_j^r$ is zero. In the second one we perform the opposite operation, by increasing $\bar{y}_j^r$ and decreasing $\bar{y}_i^r$. Both columns have one fractional coefficient less than the starting column. Furthermore, if $\pi_i \geq \pi_j$, the first column has no worse reduced cost than the starting one; the same holds for the second column if $\pi_i < \pi_j$. Therefore either the solution is not optimal (which contradicts the assumption) or an alternative optimal solution can be found by substituting the starting column with a new one. The same argument can be applied to columns containing any number of fractional coefficients $\bar{y}_i^r$.

Hence, let $\bar{\Omega}_k \subseteq \Omega_k$ be the set of extreme points having at most one fractional coordinate. We can prove the following slightly stronger result.

Corollary 2. *Any optimal column in Ω_k can be obtained as a convex combination of points in $\bar{\Omega}_k$.*

The proof is omitted. Dynamically generating columns of this model requires the solution of a resource constrained elementary shortest path problem, with the complicating feature that

prizes depend on the served fraction of each customer demand; it shares some features with the pricing problem described in Liberatore et al. (to appear), which is known to be difficult to solve. In order to obtain a more tractable relaxation, we operate as follows: let be $a_i^r = \lceil \bar{y}_i^r \rceil$; that is a_i^r takes value 0 if customer i is not visited in route r , and value 1 if it is visited, and either served completely or fractionally.

First, for each $k \in K$ we define $\tilde{\Omega}_k$ as the set of columns satisfying the following property:

$$\sum_{i \in V | a_i^r = 1} d_i - \max_{i \in V | a_i^r = 1} d_i + 1 \leq Q$$

and we observe that $\bar{\Omega}_k \subseteq \tilde{\Omega}_k$. In fact, in order to obtain the smallest capacity consumption, the customer having the largest demand is the split one, and only a very small fraction of its demand is served. Since we are assuming data is integer, the smallest fraction of demand is always 1.

Then we substitute each constraint of the set (14) as follows:

$$\sum_{k \in K} \sum_{r \in \tilde{\Omega}_k} a_i^r \lambda_r \geq 1 \quad \forall i \in V \quad (17)$$

since we are rounding up the left hand side coefficients of greater than or equal constraints, this yields a relaxation of model (13) – (16).

Furthermore, for each $k \in K$ we define $y_{ik} = \sum_{r \in \tilde{\Omega}_k} a_i^r \lambda_r$, we disaggregate each constraint of the set (17) as $\sum_{r \in \tilde{\Omega}_k} a_i^r \lambda_r = y_{ik}$ and we impose that $\sum_{k \in K} y_{ik} = 1$ for all $i \in V$, without affecting the feasibility of any solution. As in model (1)–(8), each term y_{ik} still represents the fraction of demand of customer i served by vehicle k .

Finally, we enforce inequalities (4) in the model. In fact, by relaxing constraints (14) with (17) the amount of demand that can be served by each route may exceed the vehicle capacity.

The model obtained in this way, with few slight simplifications, is presented hereafter:

$$\min z_{MP} = \sum_{k \in K} \sum_{r \in \tilde{\Omega}_k} c_r \lambda_r \quad (18)$$

$$\text{s.t. } \sum_{r \in \tilde{\Omega}_k} a_i^r \lambda_r \geq y_{ik} \quad \forall i \in V, k \in K \quad (\gamma_{ik}) \quad (19)$$

$$\sum_{r \in \tilde{\Omega}_k} \lambda_r \leq 1 \quad \forall k \in K \quad (\delta_k) \quad (20)$$

$$\sum_{k \in K} y_{ik} = 1 \quad \forall i \in V \quad (21)$$

$$\sum_{i \in V} d_i y_{ik} \leq Q \quad \forall k \in K \quad (22)$$

$$\lambda_r \geq 0 \quad \forall k \in K, r \in \tilde{\Omega}_k \quad (23)$$

$$y_{ik} \geq 0 \quad \forall i \in V, k \in K \quad (24)$$

This model contains two types of variables: λ_r variables which are exponential in number, and whose corresponding columns encode feasible routes for each vehicle, and y_{ik} variables, which are polynomial in number, and which represent the fractional of demand assignments. Only constraints (19), which are still polynomial in number, link the two types of variables: each of them impose that no demand of customer i is served by vehicle k unless some route is selected for vehicle k which visits i . Constraints (21) and (22) are analogous to constraints (3) and (4) in the flow-based formulation. The overall model has therefore an exponential number of (route) variables, but a polynomial number of constraints. We report between parenthesis, the dual variables associated with covering and convexity constraints (19) and (20), respectively. When vehicles are identical, symmetry problems might arise. Therefore, we add to this model the symmetry breaking cuts described by Jin et al. (2007). We omit the description of these cuts, since they do not substantially change the model or the column generation algorithm.

4 Solving the model

We optimize model (18)–(24) exploiting column generation techniques (Desaulniers et al., 2005). We start with a restricted problem containing all the constraints, the y_{ik} variables and, for each $k \in K$ and each $i \in V$ the column encoding the route in which vehicle k starts from the depot, serves customer i only and returns to the depot. This ensures the restricted problem has always a feasible solution. Then we dynamically search for the most negative reduced cost column for each $k \in K$ (that is, we solve $|K|$ pricing problems). Any such column is inserted in the restricted problem, which is then optimized, until no more negative reduced cost columns are found. In this way we obtain an optimal solution to (18)–(24).

For each $k \in K$ the reduced cost of a column r can be computed as follows:

$$\tilde{c}_r = c_r - \sum_{i \in V} a_i^r \gamma_{ik} + \delta_k$$

The pricing problem can be modeled as a Resource Constrained Elementary Shortest Path Problem (RCESPP) formulated on a network in which both the source and the sink nodes s and t of the path represent the depot 0. Dual prizes and penalties are taken into account in a modified cost structure related to the network: each vertex has a positive prize contribution given by γ_{ik} . The dual penalty δ_k is a constant and thus not considered during the optimization phase but taken into account when computing the optimal reduced cost. Vertex prizes can be alternatively spread over the arcs as follows:

$$\tilde{c}_{ij} = c_{ij} - \frac{1}{2}\gamma_{ik} - \frac{1}{2}\gamma_{jk}$$

In this context the RCESPP turns out to be potentially modeled on a network with negative cost cycles. In such situation the elementarity of the path cannot be ensured by cost minimization but must be enforced. To this purpose we adapt the Decremental State Space Relaxation algorithm recently proposed by Righini and Salani (to appear).

The algorithm iteratively extends partial paths ending at a node in all feasible ways appending outgoing arcs. States are associated with each vertex i and represent partial paths from the source s to the vertex i . With each state of node i are associated a resource consumption vector and a cost. To encode the constraints of the pricing problem we use five resources:

- L is a vector of binary components, representing the visited customers. L_j equals 1 if customer j is visited in the partial path, 0 otherwise,
- U is a vector of binary components, representing the set of unreachable customers. This technique, proposed by Feillet et al. (2004), allows to strengthen the domination criteria. A vertex j is unreachable from a state S when its demand would violate the capacity or when its time window would be violated by any path originated from state S ,
- t_i represents the instant in which the visit at customer i begins,
- D is the sum of the customers' demands visited in the partial path,
- d_{sc} is the demand of the potential split customer in the partial path,
- C is the cost of the partial path.

Extension. States are iteratively extended to form a complete path. An extension of a state (L, U, t_i, D, d_{sc}) associated with a node i is feasible through the arc (i, j) if and only if:

1. The j -th components of the vectors L and U equal 0,
2. The time window in j is respected, that is $t_i + s_i + t_{ij} \leq h_j$,
3. The capacity constraint can still be respected, that is $D + d_j - \max(d_j, d_{sc}) + 1 \leq Q$

A new state is created at node j and resources are update accordingly:

- $L_j = 1$
- $U_k = 1 \ \forall k \mid L_k = 0$ and an extension to k violates conditions 2. and 3.,
- $t_j = \max(g_j, t_i + s_i + t_{ij})$,
- $D_j = D_i + d_j$
- $d_{sc} = \begin{cases} d_j & \text{if } d_j > d_{sc} \\ d_{sc} & \text{otherwise} \end{cases}$,
- $C_j = C_i + \tilde{c}_{ij}$.

Dominance The efficiency of the algorithm depends on the number of generated states. It is essential to keep feasible states that could lead to optimal paths and fathom the others. Domination tests are performed for this purpose. We say that a state $S' = (L', U', t', D', d'_{sc})$ dominates a state $S'' = (L'', U'', t'', D'', d''_{sc})$ only if:

$$L' + U' \leq L'' + U'' \quad (25)$$

$$t' \leq t'' \quad (26)$$

$$D' \leq D'' \quad (27)$$

$$D' - d'_{sc} \leq D'' - d''_{sc} \quad (28)$$

$$C' \leq C'' \quad (29)$$

and at least one inequality is strict.

Conditions (27) and (28) are sufficient for dominance although other sufficient conditions exist.

5 Comparing the bounds

In this section we establish dominance relations between several bound. We consider the LP relaxation bound of the flow-based formulation (1) – (8) (FLP in the remainder), the bound given by (13) – (16) (CCLP) and the bound given by (18) – (24) (MP). Moreover, let us indicate as NCLP the lower bound obtained by optimizing the LP problem obtained by Dantzig Wolfe decomposition of model (1) – (8), when only constraints (2) and (6) are considered during convexification.

Observation 1 *The NCLP bound can be obtained by setting $\tilde{\Omega}_k = \Omega_k$ for each $k \in K$ in model (18) – (24).*

Proof. The result can be obtained by explicitly performing a Dantzig-Wolfe reformulation of model (18) – (24) in which constraints (2) and (6) are convexified. Such steps are omitted.

Observation 2 *The CCLP bound dominates the MP bound, which in turn dominates the NCLP bound.*

Proof. By construction, model (18) – (24) is a relaxation of model (13) – (16); in the meanwhile, the MP bound is obtained as the NCLP bound, but optimizing over a smaller set of columns.

Observation 3 *The CCLP bound dominates the NCLP bound, which in turn dominates the FLP bound.*

Proof. The set of feasible solutions in the CCLP bound is a subset of the feasible solutions in the NCLP bound. In fact, in the Dantzig-Wolfe reformulation for CCLP also constraints (4) are convexified, and therefore a larger region has been replaced by the convex hull of its integer points. The same holds between the NCLP bound and the FLP bound.

Finally, we consider the formulation for the SDVRP introduced by Gendreau et al. (2006). We report their model using the same notation introduced in this paper:

$$\min z_{GDFG} = \sum_{k \in K} \sum_{r \in \Omega_k} c_r \lambda_r \quad (30)$$

$$s.t. \sum_{k \in K} \sum_{r \in \Omega_k} a_i^r y_i^r \geq 1 \quad \forall i \in V \quad (31)$$

$$Q_r \lambda_r \geq \sum_{i \in V} a_i^r d_i y_i^r \quad \forall k \in K, \forall r \in \Omega_k \quad (32)$$

$$a_i^r \min\{Q_r, d_i\} \lambda_r \geq a_i^r d_i y_i^r \quad \forall i \in V, \forall k \in K, \forall r \in \Omega_k \quad (33)$$

$$\sum_{r \in \Omega_k} \lambda_r \leq 1 \quad \forall k \in K \quad (34)$$

$$\lambda_r \in \{0, 1\}, \quad y_i^r \geq 0 \quad \forall k \in K, \forall i \in V, \forall r \in \Omega_k \quad (35)$$

Each $r \in \Omega_k$ represents the index of a column encoding a feasible route for vehicle $k \in K$, c_r is the cost of such route, each term a_i^r is 1 if customer i is visited in route r (0 otherwise) and Q_r is the minimum between Q and $\sum_{i \in V} d_i a_i^r$, that is an upper bound on the amount of demand which can be served using route r . Each binary variable λ_r indicated whether route r is selected or not, while each continuous variable y_i^r represents the fraction of the demand of customer i which is assigned to route r . Constraints (31) impose that each customer is fully served, constraints (32) impose that the fraction of demands assigned to each route do not exceed route capacity, constraints (33) ensure no demand is served when a route is not selected and constraints (34) impose that at most one route is selected for each vehicle. When vehicles are identical, this model allows to solve symmetry problems by surrogating constraints (34) and considering a unique set of routes $\Omega = \bigcup_{k \in K} \Omega_k$. The column generation method applied to this model has three main issues: first, for any generated route variable λ_r an additional set of y_i^r flow variables, one for any customer which is visited in route r has to be added to the problem, while the number of flow variables in our model is polynomial and they do not need dynamic generation; second, the number of constraints (32) and (33) grows exponentially with problem dimension, and needs to be generated *during* the column generation process in order to be effective, while our model involves a polynomial number of constraints; third, the reduced cost of a column depends also on the coefficient Q_r , that is it depends on the overall served demand: as reported by Gendreau et al. (2006) this yields a difficult pricing problem, while in our case such subproblem is a well understood RCESPP.

We indicate as GDFG the lower bound obtained by the linear relaxation of this model.

Proposition 2. *The GDFG bound dominates the NCLP bound.*

Proof. By setting $y_{ik} = \sum_{r \in \Omega_k} y_i^r$ it is easy to see that each solution that satisfies constraints (31) also satisfies constraints (19).

Then, we show that each solution that satisfies constraints (32) also satisfies constraints (22). In fact, by surrogating these constraints we obtain, for each $k \in K$

$$\sum_{r \in \Omega_k} Q_r \lambda_r \geq \sum_{r \in \Omega_k} \sum_{i \in V} a_i^r d_i y_i^r$$

Since constraints (34) imply $\sum_{r \in \Omega_k} \lambda_r \leq 1$ and since $Q_r \leq Q$ the following inequalities hold:

$$Q \geq \sum_{i \in V} d_i \sum_{r \in \Omega_k} a_i^r y_i^r.$$

Now, constraints (31) impose a lower bound of $d_i \cdot 1$ to each term of the outer summation, but due to constraints (21) $d_i \cdot 1$ is an upper bound to $d_i \cdot y_{ik}$. This verifies that inequality $Q \geq \sum_{i \in V} d_i y_{ik}$ holds.

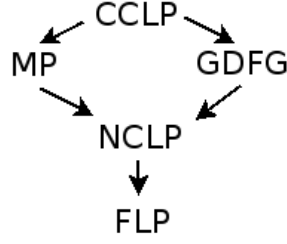


Fig. 2. Relations between bounds.

Finally, we show that each solution that satisfies constraints (33) also satisfies constraints (19). Once again, we consider the surrogate constraints $\sum_{r \in \Omega_k} a_i^r \min\{Q_r, d_i\} \lambda_r \geq \sum_{r \in \Omega_k} a_i^r d_i y_i^r$ for each $k \in K$ and for each $i \in V$, and we observe that $\sum_{r \in \Omega_k} a_i^r \min\{Q_r, d_i\} \lambda_r \leq \min\{Q, d_i\}$ and $\sum_{r \in \Omega_k} a_i^r \lambda_r \leq d_i \sum_{r \in \Omega_k} \lambda_r$. This implies $\sum_{r \in \Omega_k} \lambda_r \geq \sum_{r \in \Omega_k} a_i^r y_i^r$. Once again, 1 is a lower bound on each term of the summation to the right hand side, and 1 is an upper bound to each y_{ik} variable, and therefore inequalities $\sum_{r \in \Omega_k} \lambda_r \geq y_{ik}$ hold.

This means that the optimal solution to GDFG model is feasible for the NCLP model, but might nevertheless not be optimal.

Proposition 3. *The CCLP bound dominates the GDFG bound.*

Informally, since each column in the model for the CCLP bound represents a route together with an assignment of customer demands which do not exceed the capacity of the vehicle, any combination of columns satisfy constraints (32) and (33).

Such results can be summarized as follows: CCLP dominates MP, which dominates NCLP; in the meanwhile CCLP dominates GDFG, which dominates NCLP; however, no domination is proved between MP and GDFG.

We summarize such results in Figure 2, in which the name of each bound is included, and each arrow represents a dominance relation.

6 Computational results

Finally, we experimentally compare the quality of the MP bound with that of the GDFG bound.

Our column generation algorithm has been implemented in C and compiled using gcc 4.1.3; CPLEX 11 was used to solve the LP subproblems. The tests were run on a PC equipped with a Centrino Core 2 2GHz processor with 1GB of RAM.

We considered as benchmark the subset of 28 instances for the SDVRP with Time Windows which are solved to optimality in Gendreau et al. (2006). These were adapted from the dataset of Solomon (Solomon, 1983) by changing the vehicle capacities. These were chosen since they are the instances in which GDFG performs best, and both the GDFG bound and the IP optimum are reported. Like in the experiments conducted in their paper, we refined the bound by dynamically generating a restricted set of 2-path inequalities by (Kohl et al., 1999), involving only subsets of customers of cardinality at most 4. For the sake of comparison, we also added to the model inequalities $\sum_{k \in K} \sum_{r \in \tilde{\Omega}_k} a_i^r \lambda_r \geq \lceil d_i/Q \rceil$, imposing a lower bound on the number of visits for each $i \in V$.

We report our results in Table 1. The leftmost three columns contain the number of customers, the vehicle capacity and the name of the considered instances. In the fourth column we indicate the value of the optimum IP solution. Then, we include three blocks, each containing the value of a bound and the corresponding optimality gap. The first block refers to the GDFG bound, the second block to the MP bound, while in the third block we indicate the value of the

MP bound refined by a full generation of 2-path inequalities. In the last row of the table we report the average optimality gap of each bound. For each instance, the best bound is marked in bold. First, we remark no dominance is experimentally observed between the GDFG and the MP bound: GDFG performs better than MP when the capacity is tight, while the opposite behaviour is observed when capacity is set to 100. MP bound is very close to GDFG bound in all the instances, while being much easier to compute, and in some additional instances it allows to fully close the optimality gap at the root node.

We also compared the CPU time required to obtain the bounds when the restricted set of 2-path inequalities is generated for both the GDFG and MP bound. The CPU time reported for the GDFG bound refers to the experiments reported by Gendreau et al. (2006) using a 1.6GHz PC with 256MB of RAM and CPLEX 7.1 to solve the LP subproblems. As can be seen, the CPU time required to obtain the two bounds is quite similar on the average. However, for few instances (r103_25, r104_25, r106_25, r108_25, r112_25) MP is one order of magnitude faster than GDFG; MP seems to require a substantially higher CPU time on only one instance (r101_50).

Instance				GDFG			MP r2path			MP f2path	
V	Q	name	IP	gap	time(s)		gap	time(s)		gap	
25	30	r101_25	772.5	760.6	1.54%	0.77	752.9	2.54%	1.49	752.9	2.54%
25	30	r105_25	763.7	743.1	2.70%	3.81	721.6	5.51%	2.12	721.6	5.51%
25	50	r101_25	631.5	629.0	0.40%	0.33	631.5	0.00%	0.88	631.5	0.00%
25	50	r102_25	580.7	572.1	1.48%	1.19	555.5	4.34%	2.00	555.6	4.32%
25	50	r105_25	588.9	578.2	1.82%	0.66	567.3	3.67%	1.39	567.3	3.67%
25	50	r106_25	542.3	534.3	1.48%	2.77	519.0	4.30%	2.15	521.2	3.89%
25	50	r109_25	524.6	517.9	1.27%	1.02	503.3	4.06%	3.30	512.7	2.27%
25	50	r110_25	529.1	520.8	1.58%	2.08	501.0	5.31%	8.45	509.5	3.70%
25	50	r111_25	521.6	515.3	1.21%	3.81	498.4	4.45%	6.40	503.2	3.53%
25	100	r101_25	617.1	617.1	0.00%	0.30	617.1	0.00%	0.16	617.1	0.00%
25	100	r102_25	547.1	546.3	0.14%	0.80	547.1	0.00%	0.35	547.1	0.00%
25	100	r103_25	454.6	454.6	0.00%	8.20	454.6	0.00%	0.55	454.6	0.00%
25	100	r104_25	416.9	416.9	0.00%	16.86	416.9	0.00%	1.68	416.9	0.00%
25	100	r105_25	530.5	530.5	0.00%	0.56	530.5	0.00%	0.20	530.5	0.00%
25	100	r106_25	465.4	458.2	1.55%	2.56	465.4	0.00%	0.57	465.4	0.00%
25	100	r107_25	428.4	425.3	0.73%	7.22	424.3	0.96%	1.10	428.0	0.09%
25	100	r108_25	403.2	400.3	0.71%	48.13	397.3	1.46%	2.42	402.8	0.10%
25	100	r109_25	441.3	441.3	0.00%	1.05	441.3	0.00%	0.70	441.3	0.00%
25	100	r110_25	444.1	439.9	0.94%	2.16	439.5	1.04%	1.44	440.2	0.88%
25	100	r111_25	428.8	427.8	0.23%	4.27	427.3	0.35%	1.77	428.8	0.00%
25	100	r112_25	401.3	393.0	2.08%	12.03	392.1	2.29%	2.34	392.6	2.17%
50	100	r101_50	1044.0	1044.0	0.00%	1.94	1044.0	0.00%	24.61	1044.0	0.00%
50	100	r102_50	913.2	913.2	0.00%	64.56	912.2	0.11%	12.38	913.2	0.00%
25	100	c101_25	291.9	280.0	4.07%	1.64	279.1	4.39%	2.44	284.9	2.40%
25	100	c105_25	291.9	280.0	4.07%	1.97	276.4	5.31%	1.80	282.7	3.15%
25	100	c106_25	291.9	280.0	4.07%	1.34	279.1	4.39%	2.38	284.9	2.40%
25	100	c107_25	291.9	280.2	4.00%	2.25	276.0	5.45%	2.30	281.4	3.60%
					1.34%	7.19		2.22%	3.24		1.64%

Table 1. Comparing the bound on a subset of Solomon instances.

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